

COURSE : Data Analytics

SQL Fundamentals SELECT & FILTERING

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Due : 05/05/2026

1. Retrieve all columns

SELECT * (all)

FROM (table) employees;

∴ OUTPUT

ID	FIRSTNAME	LASTNAME	DEPARTMENT	SALARY	HIRE_DATE	CITY
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Maidoo	IT	72 000	2017-09-30	Johannesburg

2. Unique departments

SELECT DISTINCT department

FROM employees;

∴ OUTPUT

DEPARTMENT
IT
HR
Finance
Marketing

3 Names ordered by salary DESC

```
SELECT first-name, last-name  
FROM employees  
ORDER BY Salary DESC
```

∴ OUTPUT

FIRST-NAME	LAST-NAME
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4 Top 3 highest-paid employees

ID	FIRST-NAME	LAST-NAME	SALARY
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02 Filtering with WHERE

5. Employees in IT

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department = 'IT';
```

∴ OUT PUT

ID	FIRST-NAME	LAST-NAME	DEPARTMENT
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Finance with Salary > R60 000

```
SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

OUTPUT

ID	FIRST-NAME	LAST-NAME	DEPARTMENT	SALARY
3	Cathy	Zulu	Finance	65000

7. HR or Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

OUTPUT

ID	FIRST-NAME	LAST-NAME	DEPARTMENT
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Not in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department != 'IT';
```

OUTPUT	ID	FIRST-NAME	LAST-NAME	DEPARTMENT
	2	Brian	Lee	HR
	3	Cathy	Zulu	Finance
	4	David	Mokoena	Marketing

9. Using IN (IT, HR, Finance)

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

2. OUTPUT

ID	FIRST_NAME	LAST_NAME	DEPARTMENT
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. IT + Salary > 65 000 + Johannesburg

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
```

```
WHERE department = 'IT'
```

```
AND salary > 65 000
```

```
AND city = 'Johannesburg';
```

2. OUTPUT

ID	FIRST_NAME	LAST_NAME	DEPARTMENT	SALARY	CITY
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg